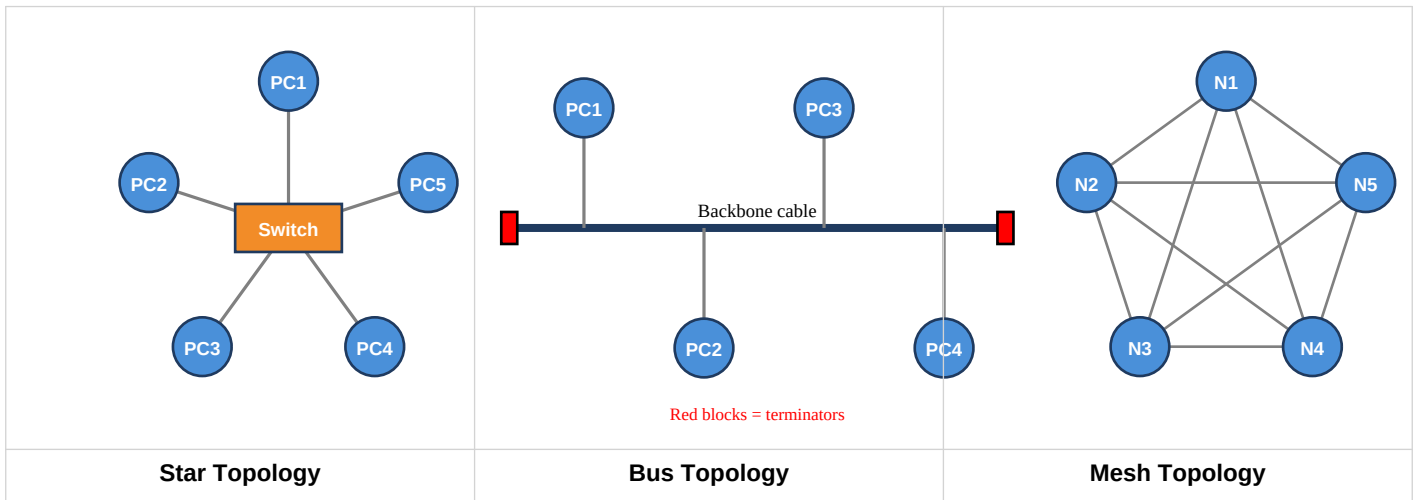


Network Topologies

Assignment Answers: Star, Bus, Ring and Mesh

Q1. Diagrams of Star, Bus and Mesh Topologies



Star: Flipkart (e-commerce)

A central server/switch handles all customer requests, so everything is easy to manage and monitor from one point, and one failing user device does not affect the others.

Bus: Small local app/LAN (e.g., a simple IPL ticket counter setup)

All devices share one backbone cable, which is cheap and enough for a few low-traffic machines that only send small amounts of data.

Mesh: WhatsApp (message routing)

Messages can travel via many alternate paths between servers, so delivery continues even if one link or node fails, giving very high reliability.

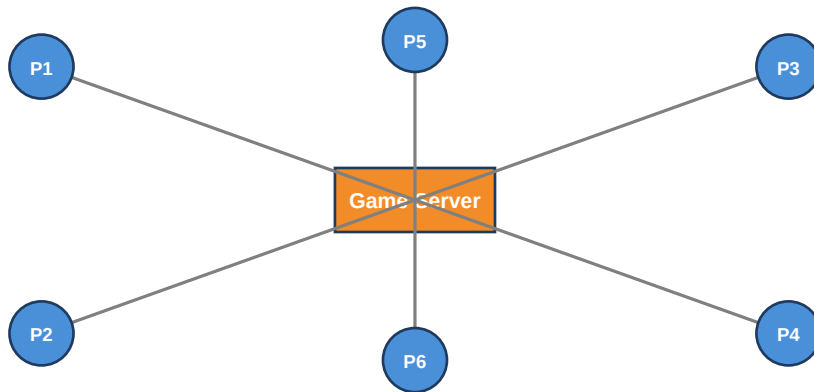
Q2. Comparison Table: Star, Bus, Ring, Mesh

Topology	Pros	Cons	Best Use Case / App Example
Star	• Easy to install and manage • Easy fault detection • One device failure does not affect others • Easy to add new devices	• Central hub is single point of failure • More cable needed • Hub/switch adds cost • Performance depends on hub capacity	Offices, schools, home Wi-Fi. Example: Zomato, where all customer, restaurant and delivery requests go through a central server.
Bus	• Cheapest, least cabling • Simple to set up • Good for small networks • Easy to extend a short distance	• Backbone failure brings down the network • Slows with more devices (collisions) • Hard to troubleshoot • Limited cable length	Small temporary labs. Example: A small local cyber cafe LAN or classroom demo network.
Ring	• Orderly, predictable data flow • No collisions • Equal access for all devices • Performs well under moderate load	• One break can stop the ring (if single ring) • Adding/removing a device disturbs the network • Hard to troubleshoot • Slower as nodes increase	Metro/telecom fiber rings. Example: Paytm's bank-to-bank backbone links that pass data in a controlled loop.
Mesh	• Very high reliability • Many alternate paths • Fast, no traffic bottleneck • Easy to isolate faulty links	• Very expensive • Complex to set up and maintain • Lots of cabling and ports • Hard to scale (links grow as $n(n-1)/2$)	Critical, always-on systems. Example: Instagram/WhatsApp data centers that interconnect heavily for zero downtime.

Q3. Network Topology for a Multiplayer Game (PUBG / Free Fire style)

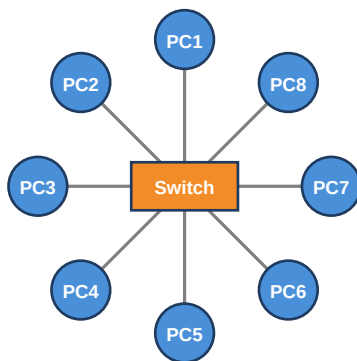
Chosen topology: Star (client-server), with servers connected to each other in a partial mesh.

Every player connects to a central game server that controls the match state, which keeps cheating low and gameplay consistent. **Speed:** region-wise servers keep ping low, and the central server syncs positions and shots quickly. **Reliability:** the backend servers are linked in a partial mesh with backups, so if one server fails, players are shifted to another. **Cost:** far cheaper than a full mesh or peer-to-peer setup because players need only one connection each and the company scales servers as demand grows.



Q4. Warehouse with 8 Computers (Flipkart-style)

Recommended topology: Star (all 8 computers connected to one central switch).



Star topology for 8 warehouse computers

Reasoning: A star network is affordable with a single 8/16-port switch, and troubleshooting is easy because a faulty cable or PC affects only that machine, so order processing never stops for everyone. It is also easy to expand: just plug new computers into the switch (or add another switch) as the warehouse grows.

Hint factors covered: Cost (low), ease of troubleshooting (high), future expansion (easy).